

Abhinav Gupta



Software & Automation Engineer | GenAI + Embedded Systems | Linux + Cybersecurity
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[GitHub](#) [LinkedIn](#) [NPM](#) [PyPI](#) [Portfolio](#)

Summary

Engineer with strong Linux foundations and cross-disciplinary expertise in software, hardware, and GenAI. Created AI-proctored platforms, real-time monitors, fixed microcontroller libraries, and a LAN screen extender for Linux. Published tools, delivered client systems, and won national hackathons.

Education

BMS Institute of Technology, Bangalore — B.E. in Information Science

Skills

- **Languages:** Python, JavaScript, Bash, Java (basic)
- **Tools:** Electron, MongoDB, Firebase, Judge0, Docker, Git
- **Embedded:** ESP32, Raspberry Pi, Arduino, ATtiny85, NRF24L01
- **Linux:** Shell scripting, chroot, debootstrap, OS automation
- **GenAI:** Prompting, OpenAI, Gemini, Web Agents, STT/TTS
- **Security:** Nmap, Metasploit, Burp Suite, Nessus, Bettercap

Major Projects

- **Prepzer0** (<https://bmsit.prepzer0.co.in>) – Developed a cutting-edge AI-driven placement testing platform with integrated AI proctoring to detect multiple faces, phone usage, and suspicious behavior. Offers HackerEarth-like experience with Judge0 backend for real-time code execution. Scalable and easy for institutions.
- **AutoCISGuard** – GUI CIS benchmark auditor for Windows and Linux with automated reporting and customizable policies using PowerShell, Bash, Python, and Electron.js. Demonstrates deep cybersecurity auditing expertise.
- **Aigle Air** – IoT air monitor measuring CO₂/O₂/PM with OLED display, providing real-time environmental feedback.
- **Elcia – Wearable Navigation Device for the Visually Impaired** – Raspberry Pi + LiDAR/ToF camera wearable that maps environment, detects obstacles, and provides voice-assisted navigation.
- **Talking Assistant Bot** – ESP32-based robotic assistant with servo motors and advanced LLM-driven TTS/STT (Google STT/Piper, Python SR), capable of lifelike conversations and emotions.

Client Work

- **Chillaxmandu.com** (<https://chillaxmandu.com>) – Full-stack e-commerce platform for liquor delivery with secure authentication, product catalog, shopping cart, and admin dashboard.
- **Halde20.ch** (<https://halde20.ch>) – Clean, responsive website for a Swiss restaurant optimized for SEO and performance.
- **LinuxScreen** – Developed Linux-compatible display extension app over LAN (Spacedesk-style) enabling multi-screen use without extra hardware.
- **LinkedIn Automation Tool** – Automated bulk messaging and connection requests on LinkedIn with custom scripts to boost outreach and networking.
- **Satellite CubeSat Project** – Miniaturized CubeSat with Arduino and NRF communication modules for satellite health monitoring and real-time ground data visualization.

Open Source & Published Work

- **Open Source Utility Libraries** – Published various NPM and PyPI packages including copy-paste utilities and workflow enhancers, contributing to developer community.
- **Application Word Count** – Lightweight desktop app for word counting with over 100+ downloads, focused on simplicity and speed.
- **AnimeWorld App** – Popular mobile/web app with 10,000+ downloads aggregating anime content with catalog, watchlists, and episode tracking using web scraping and cloud backend.
- **ATtiny85 Revival** – Fixed broken servo library for ATtiny85 microcontrollers.

Awards

1st – Pitch2Win (AI Bot)
3rd – HackDay Pondy (Aigle Air)
3rd – Pitch2Win (Expiry Biz Idea)
1st – HackSphere (Automation)
1st – Yukti (Tech Exhibition)
Consolation – Anveshna National